# Stage 4

## User Story: Interest Rate Calculator with PAN Validation and CIBIL Score

### As a user, I want to provide my PAN card number and select a loan type so that the system can generate a CIBIL score and calculate the possible interest rate based on my CIBIL score.

**1. Input**

The system should accept the following inputs using the **Scanner class**:

- PAN Card Number (String)
- Loan Type (Enum)
    - Home Loan
    - Personal Loan
    - Education Loan
    - Vehicle Loan

**2. Interest Rates**

The system should use the following interest-rate ranges:

- **Home Loan:** 6.5% – 8.5%
- **Personal Loan:** 9% – 12%
- **Education Loan:** 4% – 6%
- **Vehicle Loan:** 7% – 9%

**3. Backend Features**

**3.1 PAN Validation**

The system should validate the PAN card number.

- PAN must be exactly 10 characters long.
- It must contain 5 alphabets (A-Z), followed by 4 digits (0-9), and end with 1 alphabet (A-Z).

- If the PAN format is incorrect, display:

Invalid PAN format. Please enter a valid PAN.

### 3.2 Loan Type Validation

The system should validate that the selected loan type belongs to the predefined list:

- Home Loan
- Personal Loan
- Education Loan
- Vehicle Loan

If an invalid loan type is selected, display:

Invalid loan type selected.

### 3.3 CIBIL Score Generation

- Generate a random 3-digit CIBIL score between **600 and 900** when the user submits their PAN.
- Store the generated CIBIL score in the MySQL database using JDBC.

### 3.4 Interest Rate Calculation Based on CIBIL Score

- Calculate the interest rate based on the generated CIBIL score.
- The interest rate should be adjusted according to the CIBIL score and selected loan type.

## 4. Additional Database Fields

Add the following fields to the **User table**:

- PAN Card (String)
- CIBIL Score (Integer)

## 5. Example Workflow

**Input**

PAN: ABCDE1234F

Loan Type: Home Loan

**Processing**

- PAN validation is successful.
- A random CIBIL score is generated, for example, 720.

- Based on the CIBIL score and loan type, the system calculates the interest rate, for example, 7.5%.

**Output**

For PAN ABCDE1234F, with a CIBIL score of 720, the interest rate for Home Loan is 7.5%.

**6. Error Messages**

**Invalid PAN**

Invalid PAN format. Please enter a valid PAN.

**Invalid Loan Type**

Invalid loan type selected.

## User Story: CIBIL Score Check and Auto-Generated Credentials

**As a user, I want to provide my personal details and PAN card number so that the system can validate my information, automatically generate my username and password, send the credentials to my email, and generate my CIBIL score.**

**1. CIBIL Score Check**

**The system should accept the following details using the Scanner class:**

- **First Name (fname)**
- **Last Name (lname)**
- **Mobile Number (mobile)**
- **Email Address (email)**
- **Location (location)**
- **PAN Card (pancard)**

**Validations**

- **First Name is required and must contain only letters.**
- **Last Name is required and must contain only letters.**
- **Mobile Number is required and must contain 10 digits.**
- **Email Address is required and must follow a valid email format.**
- **Location is required.**
- **PAN Card is required and must follow the PAN format, for example: ABCDE1234F.**

**The input validation should follow the registration validation requirements.**

**If validation fails, display the appropriate error message.**

**If validation passes, proceed with username and password generation.**

**2. Auto-generate Username and Password**

**Username Generation**

**The system should automatically generate the username using:**

- **First 3 letters of First Name**
- **First 3 letters of Last Name**
- **Last 4 digits of Mobile Number**

**Example:**

**fname = Shivani**
**lname = Chaudhari**
**mobile = 9876543210**

**Generated username:**

**ShiCha3210**

**Password Generation**

**The system should randomly generate an 8–10 character password containing:**

- **Letters**
- **Numbers**
- **Symbols**

**Example:**

**@Shiv!123**

**The generated username and password should be stored in the MySQL database using JDBC.**

**The User data should contain the generated:**

- **Username**
- **Password**

### 3. Send Username and Password via Email

**After the username and password are generated and the user is created:**

- **Send an actual email to the user's registered email address.**
- **The email should contain the autogenerated username and password.**
- **The email subject should be similar to:**

**Your Username and Password for Loan Application is**

**The email should contain the generated credentials.**

**Email functionality should be implemented using a Java email library/SMTP.**

### 4. Generate and Display CIBIL Score

**After successful input validation:**

- **Generate a random CIBIL score between 600 and 900.**
- **Display the generated CIBIL score to the user through the console.**
- **Store the generated CIBIL score in the MySQL database for future reference.**

**The CIBIL score should be generated after the username and password are generated and the email containing the credentials has been sent.**

### 5. Apply for Loan

**After displaying the CIBIL score, provide the user with an Apply for Loan option in the console.**

**When the user selects this option, the system should proceed to the loan application functionality where the user can apply for a loan.**

The application should be **console-based only**. No frontend, Spring Boot, Spring MVC, or web-based implementation is required.

All database operations should be performed using **JDBC**.